1 T-Test (Student's T-Test)

Introduction

T-test is a statistical test used to determine whether there is a **significant difference between sample mean and population mean** (or between two sample means) when **population standard deviation is unknown** or **sample size is small (n < 30)**.

Why use T-Test?

- In small samples, population variance (σ^2) is usually unknown.
- Z-test cannot be applied.
- T-test uses **sample standard deviation (s)** instead.

Intuition

It measures **how far the sample mean deviates from the expected population mean** relative to sample variation.
If deviation is large → difference is significant.

Types of T-Test

1. **One-Sample T-Test**

- Compares **sample mean** to **known population mean**
- Formula:

$$t = \frac{\bar{x} - \mu}{s/\sqrt{n}}$$

Where
$\bar{x}$ = sample mean
μ = population mean
s = sample standard deviation
n = sample size

- **Example:** Check if a small class of 15 students has average marks different from 70.

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2. **Two-Sample T-Test (Independent)**

- Compares **means of two independent samples**
- Formula:

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{s_1^2/n_1 + s_2^2/n_2}}$$

- **Example:** Compare marks of two different sections of students.

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3. **Paired T-Test**

- Compares **means of two related samples**
- Example: Before and after scores of same students after a coaching program

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Graphical Interpretation

T-distribution is similar to normal distribution but **wider tails** (more variation in small samples).

Data Science Relevance

- Small dataset hypothesis testing
- A/B testing with few observations
- Checking feature impact in experiments

2️⃣ Chi-Square Test (χ^2 Test)

Introduction

Chi-Square test is used to **analyze categorical data**.

- Tests **association between two categorical variables**
- Determines whether observed frequencies differ significantly from expected frequencies.

Types of Chi-Square Test

1. **Chi-Square Goodness-of-Fit**

- Tests if a categorical variable **matches expected distribution**

Example:
A die is rolled 60 times. Are all sides equally likely?

---

2. **Chi-Square Test of Independence**

- Tests **relationship between two categorical variables**

Example:
Gender vs Choice of online/offline course

Gender	Online	Offline
Male	30	20
Female	25	25

Formula

$$\chi^2 = \sum \frac{(O_i - E_i)^2}{E_i}$$

Where
O_i = Observed frequency
E_i = Expected frequency

Data Science Relevance

- Market research (Customer preference vs demographics)
- Feature selection for categorical features
- Contingency analysis

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Python Practical (Jupyter)

```
import numpy as np
from scipy import stats

# Example: One-sample t-test
data = [65,70,72,68,75,80,60,62,78,74]
population_mean = 70

t_stat, p_value = stats.ttest_1samp(data, population_mean)
print("T Statistic:", t_stat)
print("P Value:", p_value)
```

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Topic	Data Type	Test Type	Example
T-Test	Numerical	1-sample / 2-sample / Paired	Marks small class
Chi-Square	Categorical	Goodness / Independence	Gender vs Course choice
ANOVA	Numerical, Multiple Groups	One-Way / Two-Way	Marks in 3 departments

Practical 1: Z-Test (One-Sample)

Objective:

Check whether a small sample mean differs significantly from a population mean.

Scenario:

Average battery life of phones is claimed to be **10 hours**. A sample of 15 phones is tested.